Experiment - 8

AIM -> To convert an ER Diagram into a relational Schema and implement it using MySQL.

Relational Schema =>

Customer (customer-ID, Name, Email, Phone)

PRODUCT (Product-ID, Product Name, Price, Category-ID)

Category (category-ID, (ategory-Name)

ORDERS (Order-ID, ORDER-Date, (ustomer-ID)

ORDER-ITEM (ORDER-ID, Product-ID, Quality, Price)

SQL ->

```
Create TABLE CUSTOMER(
CUSTOMER-ID INT PRIMARY key,
NAME VARCHAR(50) NotNULL,
EMAIL VARCHAR(100) UNIQUE);

CREATE TABLE CATEGORY(
CATEGORY-ID INT PRIME KEY,
CATEGORY-NAME VARCHAR(50));

CREATE TABLE PRODUCT(
CATEGORY-ID INT PRIMARY KEY,
PRODUCT_NAME VARCHAR(50),
PRICE DECIMAL(10.2),
CATEGORY-ID INT,
FOREIGN Key (Category-ID)
REFERENCES CATEGORY (Category-ID));
```

```
CREATE TABLE ORDERS(
 Order_ID INT PRIMARY KEY,
 Order_Data .DATE
  Customer_ID INT,
  FOREIGN Key (customer_ID)
  References Customer (CUSTOMER_ID));

CREATE TABLE ORDER_ITEMS
  ORDER_ID INT,
  Product_ID INT,
  Quantity INT,
  Price Decimal (10, 2),
  PRIMARY Key (order_ID, Product_ID)
  FOREIGN Key (order_ID)
   REFERENCES ORDERS (ORDER_ID)
     ON DELETE CASCADE);
```

SAMPLE DATA:

```
INSERT INTO CUSTOMER VALUES
  (1, 'RAHUL', rahul@gmail.com);
INSERT INTO CATEGORY values
  (101, 'ELECTRONICS');
INSERT INTO PRODUCTVALUES
  (201, 'LAPTOP', 65000, 101);
```

VIOLATION

```
INSERT INTO ORDERS VALUES
  (501, '2026-8-19', 99);
```

Result: Foreign Key error because customer_ID 99 does not exist.